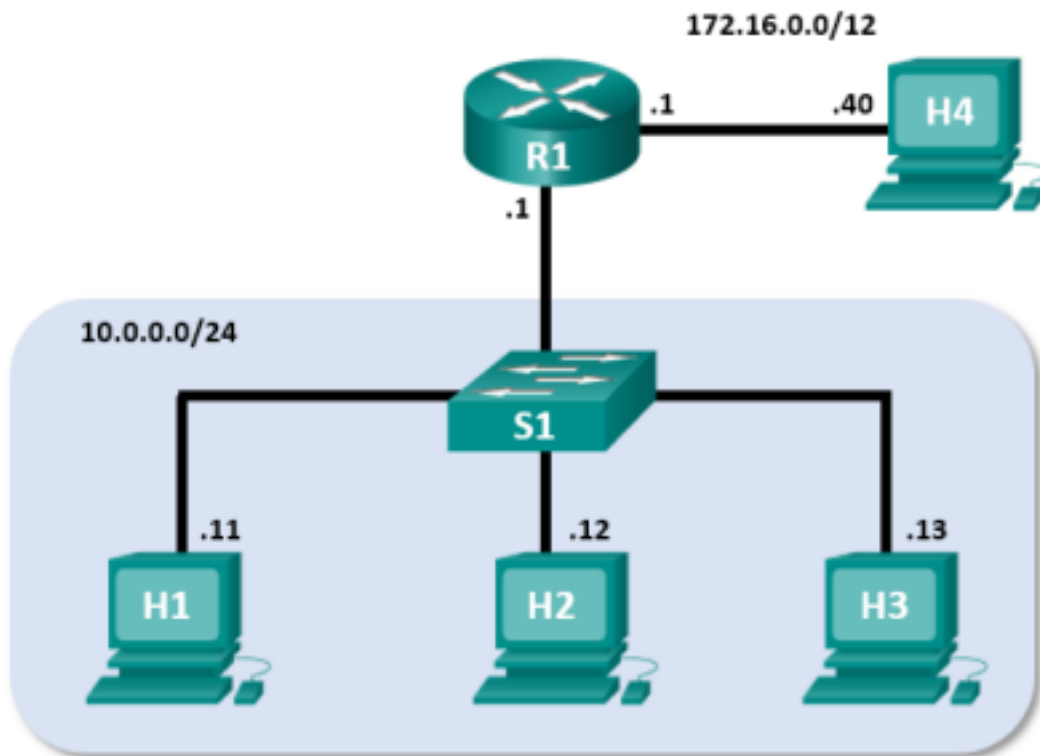


Lab - Using Wireshark to Observe the TCP 3-Way

Handshake Mininet Topology



Objectives

Part 1: Prepare the Hosts to Capture the Traffic

Part 2: Analyze the Packets using Wireshark

Part 3: View the Packets using tcpdump

Background / Scenario

In this lab, you will use Wireshark to capture and examine packets generated between the PC browser using the HyperText Transfer Protocol (HTTP) and a web server, such as www.google.com. When an application, such as HTTP or File Transfer Protocol (FTP) first starts on a host, TCP uses the three-way handshake to establish a reliable TCP session between the two hosts. For example, when a PC uses a web browser to surf the internet, a three-way handshake is initiated, and a session is established between the PC host and web server. A PC can have multiple, simultaneous, active TCP sessions with various web sites.

Required Resources

- CyberOps Workstation virtual machine

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Instructions

Part 1: Prepare the Hosts to Capture the Traffic

a. Start the CyberOps VM. Log in with username **analyst** and the password **cyberops**.

b. Start Mininet.

```
[analyst@secOps ~]$ sudo lab.support.files/scripts/cyberops_topo.py
```

c. Start host H1 and H4 in Mininet.

```
*** Starting CLI:
```

```
mininet> xterm H1
```

```
mininet> xterm H4
```

d. Start the web server on H4.

```
[root@secOps analyst]#  
/home/analyst/lab.support.files/scripts/reg_server_start.sh
```

e. For security purposes, you are not able to run Firefox from the root user account. On host H1, use the switch user command to switch from the root user to the analyst user account:

```
[root@secOps analyst]# su analyst
```

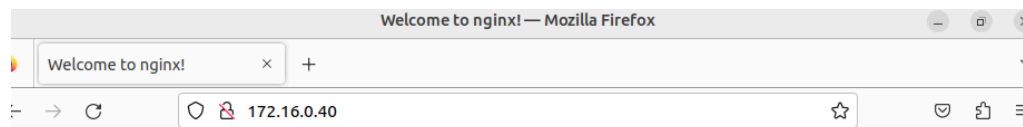
f. Start the web browser on H1. This will take a few moments.

```
[analyst@secOps ~]$ firefox &
```

g. After the Firefox window opens, start a tcpdump session in the terminal **Node: H1** and send the output to a file called **capture.pcap**. With the -v option, you can watch the progress. This capture will stop after capturing 50 packets, as it is configured with the option -c 50.

```
[analyst@secOps ~]$ sudo tcpdump -i H1-eth0 -v -c 50 -w  
/home/analyst/capture.pcap
```

h. After the tcpdump starts, quickly navigate to 172.16.0.40 in the Firefox web browser.



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

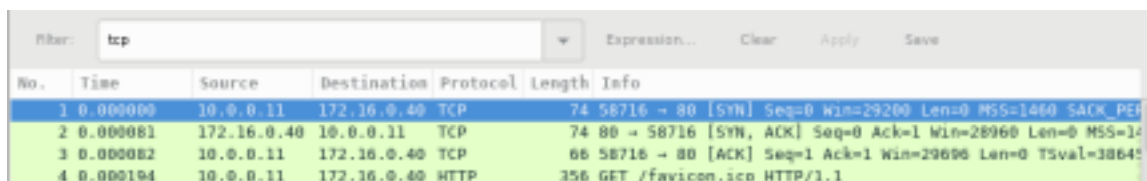
Part 2: Analyze the Packets using Wireshark

Step 1: Apply a filter to the saved capture.

- Press ENTER to see the prompt. Start Wireshark on **Node: H1**. Click **OK** when prompted by the warning regarding running Wireshark as superuser.

```
[analyst@secOps ~]$ wireshark &
```

- In Wireshark, click **File > Open**. Select the saved pcap file located at /home/analyst/capture.pcap.
- Apply a **tcp** filter to the capture. In this example, the first 3 frames are the interested traffic.



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.0.11	172.16.0.40	TCP	74	58716 → 80 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERFECT
2	0.000081	172.16.0.40	10.0.0.11	TCP	74	80 → 58716 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460
3	0.000082	10.0.0.11	172.16.0.40	TCP	66	58716 → 80 [ACK] Seq=1 Ack=1 Win=29696 Len=0 TSval=38645
4	0.000194	10.0.0.11	172.16.0.40	HTTP	356	GET /favicon.ico HTTP/1.1

Step 2: Examine the information within packets including IP addresses, TCP port numbers, and TCP control flags.

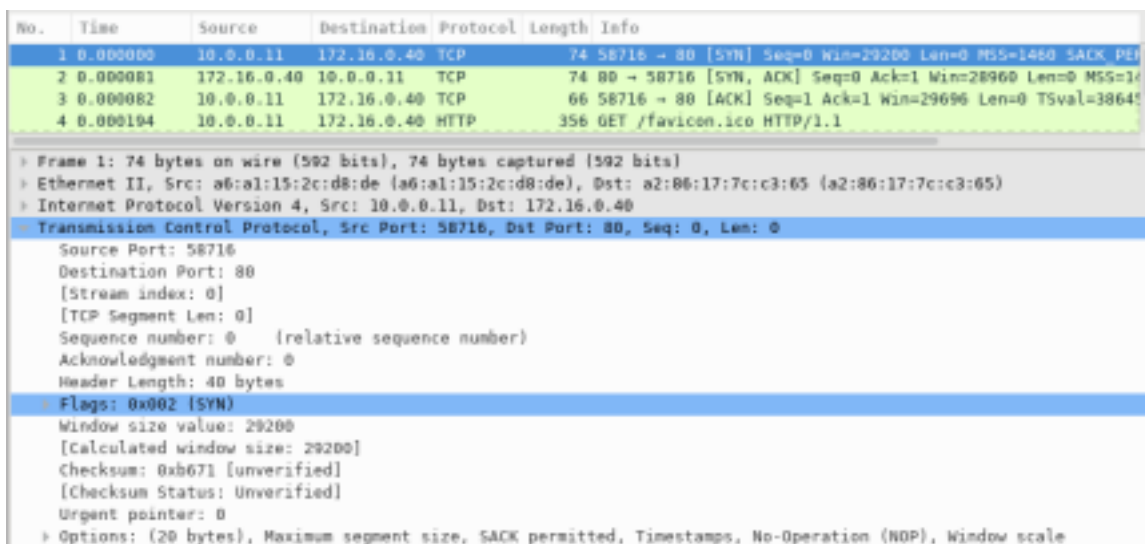
- In this example, frame 1 is the start of the three-way handshake between the PC and the server on H4. In the packet list pane (top section of the main window), select the first packet, if necessary.

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- Click the **arrow** to the left of the Transmission Control Protocol in the packet details pane to expand it and examine the TCP information. Locate the source and destination port information.
- Click the **arrow** to the left of the Flags. A value of 1 means that flag is set. Locate the flag that is set in this packet.

Note: You may have to adjust the top and middle windows sizes within Wireshark to display the necessary information.



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.0.11	172.16.0.40	TCP	74	58716 → 80 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERFECT
2	0.000081	172.16.0.40	10.0.0.11	TCP	74	80 → 58716 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460
3	0.000082	10.0.0.11	172.16.0.40	TCP	66	58716 → 80 [ACK] Seq=1 Ack=1 Win=29696 Len=0 TSval=38645
4	0.000194	10.0.0.11	172.16.0.40	HTTP	356	GET /favicon.ico HTTP/1.1

Frame 1: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface	
Ethernet II, Src: a6:a1:15:2c:d8:de (a6:a1:15:2c:d8:de), Dst: a2:86:17:7c:c3:65 (a2:86:17:7c:c3:65)	
Internet Protocol Version 4, Src: 10.0.0.11, Dst: 172.16.0.40	
Transmission Control Protocol, Src Port: 58716, Dst Port: 80, Seq: 0, Len: 0	
Source Port: 58716	
Destination Port: 80	
[Stream index: 0]	
[TCP Segment Len: 0]	
Sequence number: 0 (relative sequence number)	
Acknowledgment number: 0	
Header Length: 40 bytes	
Flags: 0x002 [SYN]	
Window size value: 29200	
[Calculated window size: 29200]	
Checksum: 0xb671 [unverified]	
[Checksum Status: Unverified]	
Urgent pointer: 0	
Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale	

What is the TCP source port number?

51460

How would you classify the source port?

Dynamic / Ephemeral / Private Port

What is the TCP destination port number?

80

How would you classify the destination port?

Well-Known Port

Which flag (or flags) is set?

SYN

What is the relative sequence number set to?

0

```
Source Port: 51460
Destination Port: 80
[Stream index: 0]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 0      (relative sequence number)
Sequence Number (raw): 3064569840
[Next Sequence Number: 1      (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1010 .... = Header Length: 40 bytes (10)
```

Lab - Using Wireshark to Observe the TCP 3-Way Handshake

- d. Select the next packet in the three-way handshake. In this example, this is frame 2. This is the web server replying to the initial request to start a session.

Filter: tcp		Expression...		Clear	Apply	Save
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.0.11	172.16.0.40	TCP	74	58716 → 80 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERF
2	0.000081	172.16.0.40	10.0.0.11	TCP	74	80 → 58716 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=14
3	0.000082	10.0.0.11	172.16.0.40	TCP	66	58716 → 80 [ACK] Seq=1 Ack=1 Win=29696 Len=0 TSval=3864
4	0.000194	10.0.0.11	172.16.0.40	HTTP	356	GET /favicon.ico HTTP/1.1

▶ Frame 2: 74 bytes on wire (592 bits), 74 bytes captured (592 bits)
 ▶ Ethernet II, Src: a2:86:17:7c:c3:65 (a2:86:17:7c:c3:65), Dst: a6:a1:15:2c:d8:de (a6:a1:15:2c:d8:de)
 ▶ Internet Protocol Version 4, Src: 172.16.0.40, Dst: 10.0.0.11
 ▶ Transmission Control Protocol, Src Port: 80, Dst Port: 58716, Seq: 0, Ack: 1, Len: 0
 Source Port: 80
 Destination Port: 58716
 [Stream index: 0]
 [TCP Segment Len: 0]
 Sequence number: 0 (relative sequence number)
 Acknowledgment number: 1 (relative ack number)
 Header Length: 40 bytes
 Flags: 0x012 [SYN, ACK]
 Window size value: 28960
 [Calculated window size: 28960]
 Checksum: 0xc85a [unverified]
 [Checksum Status: Unverified]
 Urgent pointer: 0
 Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale

What are the values of the source and destination ports?

Source port: 80

Destination port: 51460

Which flags are set?

Cò SYN và ACK (SYN-ACK).

What are the relative sequence and acknowledgment numbers set to?

Relative Sequence number: 0

Relative Acknowledgment number: 1

e. Finally, select the third packet in the three-way handshake.

Filter: tcp		Expression...		Clear	Apply	Save
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.0.11	172.16.0.40	TCP	74	58716 → 80 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERF
2	0.000081	172.16.0.40	10.0.0.11	TCP	74	80 → 58716 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=14
3	0.000082	10.0.0.11	172.16.0.40	TCP	66	58716 → 80 [ACK] Seq=1 Ack=1 Win=29696 Len=0 TSval=3864
4	0.000194	10.0.0.11	172.16.0.40	HTTP	356	GET /favicon.ico HTTP/1.1

▶ Frame 3: 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
 ▶ Ethernet II, Src: a6:a1:15:2c:d8:de (a6:a1:15:2c:d8:de), Dst: a2:86:17:7c:c3:65 (a2:86:17:7c:c3:65)
 ▶ Internet Protocol Version 4, Src: 10.0.0.11, Dst: 172.16.0.40
 ▶ Transmission Control Protocol, Src Port: 58716, Dst Port: 80, Seq: 1, Ack: 1, Len: 0
 Source Port: 58716
 Destination Port: 80
 [Stream index: 0]
 [TCP Segment Len: 0]
 Sequence number: 1 (relative sequence number)
 Acknowledgment number: 1 (relative ack number)
 Header Length: 32 bytes
 Flags: 0x010 [ACK]
 Window size value: 58
 [Calculated window size: 29696]
 [Window size scaling factor: 512]
 Checksum: 0xb669 [unverified]
 [Checksum Status: Unverified]
 Urgent pointer: 0

Examine the third and final packet of the handshake.

Which flag (or flags) is set?

Cờ **ACK** (Acknowledgment).

The relative sequence and acknowledgment numbers are set to 1 as a starting point. The TCP connection is established and communication between the source computer and the web server can begin.

Part 3: View the packets using tcpdump

You can also view the pcap file and filter for the desired information.

- a. Open a new terminal window, enter **man tcpdump**. **Note:** You may need to press ENTER to see the prompt.

Using the manual pages available with the Linux operating system, you read or search through the manual pages for options for selecting the desired information from the pcap file.

```
[analyst@secOps ~]$ man tcpdump
TCPDUMP(1) General Commands Manual TCPDUMP(1)

NAME
    tcpdump - dump traffic on a network

SYNOPSIS
    tcpdump [ -AbdDefhHIJKlLnNOpqStuUvxX# ] [ -B buffer_size ]
    [ -c count ]
    [ -C file_size ] [ -G rotate_seconds ] [ -F file ]
    [ -i interface ] [ -j tstamp_type ] [ -m module ] [ -M secret ] [ --number ]
    [ -Q in|out|inout ]
    [ -r file ] [ -V file ] [ -s snaplen ] [ -T type ] [ -w file ] [ -W
filecount ]
    [ -E spi@ipaddr algo:secret,... ]
    [ -y datalinktype ] [ -z postrotate-command ] [ -Z user ] [
--time-stamp-precision=tstamp_precision ]
    [ --immediate-mode ] [ --version ]
    [ expression ]
<some output omitted>
```

To search through the man pages, you can use / (searching forward) or ? (searching backward) to find specific terms, and **n** to forward to the next match and **q** to quit. For example, search for the information on the switch **-r**, type **/-r**. Type **n** to move to the next match.

What does the switch **-r** do?

Tùy chọn **-r** (read) cho phép **tcpdump** đọc và hiển thị các gói tin từ một tệp tin lưu trữ (tệp **.pcap**) thay vì bắt lưu lượng trực tiếp từ giao diện mạng thực tế.

- b. In the same terminal, open the capture file using the following command to view the first 3 TCP packets captured:

```
[analyst@secOps ~]$ tcpdump -r /home/analyst/capture.pcap tcp -c 3
reading from file capture.pcap, link-type EN10MB (Ethernet)
13:58:30.647462 IP 10.0.0.11.58716 > 172.16.0.40.http: Flags [S], seq 2432755549, win
29200, options [mss 1460,sackOK,TS val 3864513189 ecr 0,nop,wscale 9], length 0
```

```
13:58:30.647543 IP 172.16.0.40.http > 10.0.0.11.58716: Flags [S.], seq 1766419191, ack 2432755550, win 28960, options [mss 1460,sackOK,TS val 50557410 ecr 3864513189,nop,wscale 9], length 0
13:58:30.647544 IP 10.0.0.11.58716 > 172.16.0.40.http: Flags [.], ack 1, win 58, options [nop,nop,TS val 3864513189 ecr 50557410], length 0
```

To view the 3-way handshake, you may need to increase the number of lines after the **-c** option.

- c. Navigate to the terminal used to start Mininet. Terminate the Mininet by entering quit in the main CyberOps VM terminal window.

```
mininet> quit
*** Stopping 0 controllers

*** Stopping 2 terms
*** Stopping 5 links
.....
*** Stopping 1 switches
s1
*** Stopping 5 hosts
H1 H2 H3 H4 R1
*** Done
[analyst@secOps ~]$
```

- d. After quitting Mininet, enter **sudo mn -c** to clean up the processes started by Mininet. Enter the password **cyberops** when prompted.

```
[analyst@secOps ~]$ sudo mn -c
[sudo] password for analyst:
```

Reflection Questions

1. There are hundreds of filters available in Wireshark. A large network could have numerous filters and many different types of traffic. List three filters that might be useful to a network administrator.
2. What other ways could Wireshark be used in a production network?

